

Relation Algebra

§1. Notation

Traditionally Adjunction $F \dashv G$ is defined as $F(x) \leq y$ iff $x \leq G(y)$, and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

By define $X : * \mapsto x$, $Y : * \mapsto y$ and use diagram order, application can be replaced by composition: $XF \leq Y$ iff $X \leq YG$

(1a)

¹ • write F as $/$, G as $\backslash$

² • render $a \leq b$ as $\boxed{\frac{a}{b}}$

³ • **rewrite** $X \leq XF$ as $1 \leq F$

$$\begin{aligned} \boxed{\frac{X/}{Y}} &= \boxed{\frac{X}{Y\backslash}} && \text{adj} \\ \\ \boxed{\frac{X/}{X/}} &\xRightarrow{\text{adj}} \boxed{\frac{X}{X\wedge}} \xRightarrow{\text{rewrite}} \boxed{\frac{*}{\wedge}} \xRightarrow{\text{unit}} \begin{array}{c} \xRightarrow{\text{put } \backslash \text{ on left}} \boxed{\frac{\backslash}{\wedge\backslash}} \\ \xRightarrow{\text{put } / \text{ on right}} \boxed{\frac{/}{\wedge/}} \end{array} \\ \\ \boxed{\frac{Y\backslash}{Y\backslash}} &\xRightarrow{\text{adj}} \boxed{\frac{Y\vee}{Y}} \xRightarrow{\text{rewrite}} \boxed{\frac{\vee}{*}} \xRightarrow{\text{counit}} \begin{array}{c} \xRightarrow{\text{put } / \text{ on left}} \boxed{\frac{\vee/}{/}} \\ \xRightarrow{\text{put } \backslash \text{ on right}} \boxed{\frac{\vee\backslash}{\backslash}} \end{array} \end{aligned}$$

now we have $\vee/ = /$ and $\vee\backslash = \backslash$, and you just discovered string diagrams.

§1.1. Adjunctions

(1.1a)

	F ⊣ G	F's type	monad FG	1 ⊆ FG	GF ⊆ 1	F preserves u	G preserves n	FGF=F	GFG=G
1	$\triangleleft \dashv \triangleright$	$A \rightarrow A \otimes A$	$\triangleleft \triangleright = 1$	$1 \subseteq \triangleleft \triangleright$	$\triangleright \triangleleft \subseteq 1$	$(R \cup S) \triangleleft = R \triangleleft \cup S \triangleleft$	$(R \cap S) \triangleright = R \triangleright \cap S \triangleright$	$\triangleleft \triangleright \triangleleft = \triangleleft$	$\triangleright \triangleleft \triangleright = \triangleright$
2	$\multimap \dashv \multimap$	$A \rightarrow I$	$\multimap \multimap = T$	$1 \subseteq \multimap \multimap$	$\multimap \multimap \subseteq 1$	$(R \cup S) \multimap = R \multimap \cup S \multimap$	$(R \cap S) \multimap = R \multimap \cap S \multimap$	$\multimap \multimap \multimap = \multimap$	$\multimap \multimap \multimap = \multimap$
3	$\circ \dashv \circ$	$(A \rightarrow B) \rightarrow (B \rightarrow A)$	$R^{\circ \circ} = R$	$R^{\circ \circ} = R$	$R^{\circ \circ} = R$	$(R \cup S)^{\circ} = S^{\circ} \cup R^{\circ}$	$(R \cap S)^{\circ} = R^{\circ} \cap S^{\circ}$	$R^{\circ \circ} = R$	$R^{\circ \circ} = R$
4	$\circ \triangleleft \dashv \triangleright \multimap$	$(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$	1	$(\circ \triangleleft \otimes 1) = 1$ $(1 \otimes \triangleright \multimap) = 1$	$(1 \otimes \circ \triangleleft) = 1$ $(\triangleright \multimap \otimes 1) = 1$	=	=	=	=
5	$\triangleright \multimap \dashv \triangleleft \circ$	$(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$	1	$(1 \otimes \circ \triangleleft) = 1$ $(\triangleright \multimap \otimes 1) = 1$	$(\circ \triangleleft \otimes 1) = 1$ $(1 \otimes \triangleright \multimap) = 1$	=	=	=	=
6	$\Delta \dashv \nabla$	$(A \rightarrow B) \rightarrow (A \rightarrow B)^2$	$R \cap R = R$	$R \cap R = R$	$R \cap S \subseteq R$	$\Delta(R \cup S) = \Delta R \cup \Delta S$	$(R \cap T) \cap (S \cap U) = (R \cap S) \cap (T \cap U)$	$R \cap R = R$	$R \cap R = R$
7	$U \dashv \Delta$	$(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$	$(R, S) \mapsto (R \cup S, R \cup S)$	$R \subseteq R \cup S$	$R \cup R = R$	$(R \cup T) \cup (S \cup U) = (R \cup S) \cup (T \cup U)$	$\Delta(R \cap S) = \Delta R \cap \Delta S$	$R \cup R = R$	$R \cup R = R$
8	$\emptyset \dashv !$	$\{*\} \rightarrow (A \rightarrow B)$			$\emptyset \subseteq R$				
9	$\cdot S \dashv / S$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	S/S	$R \subseteq (RS)/S$	$(T/S)S \subseteq T$	$(S \cup T)R = (SR) \cup (TR)$	$(R \cap R') / S = (R / S) \cap (R' / S)$	$((RS)/S)S = RS$	$((T/S)S)/S = T/S$
10	$S \cdot \dashv S \setminus$	$(B \rightarrow C) \rightarrow (A \rightarrow C)$	$S \setminus S$	$R \subseteq S \setminus (SR)$	$S(S \setminus T) \subseteq T$	$R(S \cup T) = (RS) \cup (RT)$	$S \setminus (R \cap R') = (S \setminus R) \cap (S \setminus R')$	$S(S \setminus (SR)) = SR$	$S \setminus (S(S \setminus T)) = S \setminus T$
11	$R \cap \dashv R \Rightarrow$	$(A \rightarrow B) \rightarrow (A \rightarrow B)$	$X \mapsto R \Rightarrow (X \cap R)$	$X \subseteq R \Rightarrow (X \cap R)$	$R \cap (R \Rightarrow Y) \subseteq Y$	$R \cap (X \cup Y) = (R \cap X) \cup (R \cap Y)$	$R \Rightarrow S \cap T = (R \Rightarrow S) \cap (R \Rightarrow T)$	$R \cap (R \Rightarrow (X \cap R)) = X \cap R$	$R \Rightarrow (R \cap (R \Rightarrow Y)) = R \Rightarrow Y$
12	$\mathcal{D} \dashv \cdot T$	$(A \rightarrow B) \rightarrow \text{Cor } A$	$R \mapsto (\mathcal{D}R)T$	$R \subseteq (\mathcal{D}R)T$	$\mathcal{D}(AT) \subseteq A$	$\mathcal{D}(R \cup S) = \mathcal{D}R \cup \mathcal{D}S$	$(A \cap B)T = AT \cap BT$	$\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$	$(\mathcal{D}(AT))T = AT$
13	$\mathcal{R} \dashv T \cdot$	$(A \rightarrow B) \rightarrow \text{Cor } B$	$R \mapsto T(\mathcal{R}R)$	$R \subseteq T(\mathcal{R}R)$	$\mathcal{R}(TA) \subseteq A$	$\mathcal{R}(R \cup S) = \mathcal{R}R \cup \mathcal{R}S$	$T(A \cap B) = TA \cap TB$	$\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$	$T(\mathcal{R}(TA)) = TA$
14	$\cdot f \dashv \cdot f^{\circ}$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	ff°	$\text{Map}(f) \Rightarrow 1 \subseteq f f^{\circ}$	$f^{\circ} f \subseteq 1$	$(S \cup T)R = (SR) \cup (TR)$	$\text{Simple}(g) \Rightarrow (X \cap Y)g^{\circ} = (Xg^{\circ}) \cap (Yg^{\circ})$	$ff^{\circ}f = f$	$f^{\circ}ff^{\circ} = f^{\circ}$
15	$f^{\circ} \cdot \dashv f \cdot$	$(A \rightarrow C) \rightarrow (B \rightarrow C)$	ff°	$\text{Map}(f) \Rightarrow 1 \subseteq f f^{\circ}$	$f^{\circ} f \subseteq 1$	$R(S \cup T) = (RS) \cup (RT)$	$f(X \cap Y) = fX \cap fY$	$ff^{\circ}f = f$	$f^{\circ}ff^{\circ} = f^{\circ}$
16	$i \dashv E$	$\text{Map} \multimap \text{Rel}$	E	$\frac{1}{\exists} : A \rightarrow PA$	$\exists : PB \rightarrow B$	—	—	$\frac{\exists}{\exists} = 1$	$\frac{R}{\exists} \exists = R$
17	$\cdot \exists \dashv \frac{\exists}{\exists}$	$\text{Map}(A, PB) \rightarrow (A \rightarrow B)$	1	$\frac{f \exists}{\exists} = f$	$\frac{R}{\exists} \exists = R$	—	—	$\frac{f \exists}{\exists} = f \exists$	$\frac{R}{\exists} \exists = R$
18	$\frac{\exists}{\exists} \dashv \cdot \exists$	$(A \rightarrow B) \rightarrow \text{Map}(A, PB)$	1	$\frac{R}{\exists} \exists = R$	$\frac{f \exists}{\exists} = f$	—	—	$\frac{\exists}{\exists} = \frac{R}{\exists}$	$\frac{f \exists}{\exists} = f \exists$

§1.2. Composing adjunctions

(1.2a)

$$\begin{array}{c}
\begin{array}{ccccc}
\boxed{\begin{array}{c} X \\ f(R/T) \end{array}} & \xLeftrightarrow{f^{\circ} \cdot \dashv f \cdot} & \boxed{\begin{array}{c} f^{\circ}X \\ R/T \end{array}} & \xLeftrightarrow{\cdot T \dashv / T} & \boxed{\begin{array}{c} (f^{\circ}X)T \\ R \end{array}} \\
\text{f a map} & & & & \\
\xLeftrightarrow{\text{associativity}} & \boxed{\begin{array}{c} f^{\circ}(XT) \\ R \end{array}} & \xLeftrightarrow{f^{\circ} \cdot \dashv f \cdot} & \boxed{\begin{array}{c} XT \\ fR \end{array}} & \xLeftrightarrow{\cdot T \dashv / T} & \boxed{\begin{array}{c} X \\ (fR)/T \end{array}} \\
& \xRightarrow{\text{indirect equality}} & \boxed{\begin{array}{c} f(R/T) \\ (fR)/T \end{array}} & & &
\end{array}$$

$$\text{Map}(f) \Rightarrow f(R/S) = (fR)/S$$

(1.2b)

	$\cdot T$	$T \cdot$	$\cdot g$	$g^\circ \cdot$	Tn	$^\circ$	$\circ \triangleleft$	Δ
1	$\cdot S$	$R / (S_1 \ S_2) = R / S_2 / S_1$	$S \setminus (R / W) = (S \setminus R) / W$	$R/(Sg) = (Rg^\circ)/S$	$\text{Map}(f) \Rightarrow f$ $(R / S) = (f R) / S$	$(Y/S)^\circ = S^\circ \setminus (Y^\circ)$	$(RS)^\wedge = R^\wedge (S \otimes 1)$	$(R \cap R') / S = (R / S) \cap (R' / S)$
2	$S \cdot$	$S \setminus (R / W) = (S \setminus R) / W$	$(S_1 \ S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$	$S \setminus (Rg^\circ) = (S \setminus R)g^\circ$	$(g^\circ S) \setminus R = S \setminus (gR)$	$(S \setminus Y)^\circ = Y^\circ / S^\circ$	$((S \otimes 1)R)^\wedge = SR^\wedge$	$S \setminus (R \cap R') = (S \setminus R) \cap (S \setminus R')$
3	$\cdot f$	$\text{Map}(f) \Rightarrow R / (f S) = (R / S) f^\circ$	$T \setminus (Rf^\circ) = (T \setminus R) f^\circ$	$(R S)^\circ = S^\circ R^\circ$		$(R S)^\circ = S^\circ R^\circ$		$\text{Simple}(g) \Rightarrow (X \cap Y) g^\circ = (X g^\circ) \cap (Y g^\circ)$
4	$f^\circ \cdot$	$\text{Map}(f) \Rightarrow f (R / S) = (f R) / S$	$(Tf^\circ) \setminus R = f(T \setminus R)$		$(R S)^\circ = S^\circ R^\circ$	$(R S)^\circ = S^\circ R^\circ$		$f(T_1 \cap T_2) = fT_1 \cap fT_2$
5	Rn	—	—	—	$R \Rightarrow S \Rightarrow T = R \cap S \Rightarrow T$	$(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$	—	$R \Rightarrow S \cap T = (R \Rightarrow S) \cap (R \Rightarrow T)$
6	$^\circ$	$(Y/T)^\circ = T^\circ \setminus (Y^\circ)$	$(T \setminus Y)^\circ = Y^\circ / T^\circ$	$(R S)^\circ = S^\circ R^\circ$	$(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$	$R^\circ \circ = R$	$(R^\wedge)^\circ = (R^\circ \otimes 1)$ $(1 \otimes \triangleright \circ)$ both bends: the $^\circ$ section's display	$(T_1 \cap T_2)^\circ = T_1^\circ \cap T_2^\circ$
7	$\circ \triangleleft$	$(RT)^\wedge = R^\wedge (T \otimes 1)$	$((T \otimes 1)R)^\wedge = TR^\wedge$	—	—	$(R^\wedge)^\circ = (R^\circ \otimes 1)$ $(1 \otimes \triangleright \circ)$ both bends: the $^\circ$ section's display	—	—
8	u	$(S \cup T) R = (S R) \cup (T R)$	$R (S \cup T) = (R S) \cup (R T)$	$(S \cup T) R = (S R) \cup (T R)$	$R (S \cup T) = (R S) \cup (R T)$	$R \cap (S \cup T) = R \cap S \cup R \cap T$	$(R \cup S)^\circ = S^\circ \cup R^\circ$	
9	$\emptyset$	$\emptyset R = \emptyset$	$R \emptyset = \emptyset$	$\emptyset R = \emptyset$	$R \emptyset = \emptyset$	$R \cap \emptyset = \emptyset$	$\emptyset^\circ = \emptyset$	

§1.3. Adjoint triples

Beyond $u \dashv \Delta \dashv n$, which §1.1 already carries as two rows, the table holds one adjoint triple with content: $\exists_f \dashv f^* \dashv \forall_f$ along a map f , once on each side of the composite. $f : A \rightarrow B$ throughout this subsection.

(1.3a)

$$\begin{aligned} \cdot f \dashv \cdot f^\circ \dashv /f^\circ \\ f^\circ \cdot \dashv f \cdot \dashv f \setminus \end{aligned}$$

The first two links of each chain are the map shunting rules, and the third is §1.1's division row read at a chosen divisor: $\cdot S \dashv /S$ at $S := f^\circ$, and $S \cdot \dashv S \setminus$ at $S := f$.

The third link is not the first over again. For a map f , $/f$ collapses to $\cdot f^\circ$; being a map is exactly what that collapse spends $\text{Map}(f) \Rightarrow R / (f S) = (R / S) f^\circ$ — f° is not a map, so $/f^\circ$ does not collapse in turn, and the three operators stay distinct. On $\text{Rel}(\mathbb{I}, A) = PA$ they are the image triple:

(1.3b)

	operator	acts by	name
1	$\cdot f$	$S \mapsto f[S]$	direct image, $\exists_f$
2	$\cdot f^\circ$	$T \mapsto f^{-1}[T]$	inverse image, f^*
3	$/f^\circ$	$S \mapsto \{b : f^{-1}(b) \subseteq S\}$	$\forall_f$

§1.1's $\mathcal{D} \dashv \cdot T$ is this same chain along the projection $A \otimes B \rightarrow A$, read through $\text{Rel}(A, B) = P(A \otimes B)$: $\mathcal{D}R = \{(a, a) : \exists b. aRb\}$, AT is the relation that ignores b altogether, and the third link is $R \mapsto 1 \cap R / T = \{(a, a) : \forall b. aRb\}$.

(1.3c)

$$\mathcal{D} \dashv \cdot T \dashv 1 \cap \cdot / T$$

Three is where it stops, and one map breaks both ends. Take $A=\{a_1, a_2\}$ and $B=\{b\}$: $f[\{a_1\}] \cap f[\{a_2\}] = \{b\}$ while $f[\{a_1\} \cap \{a_2\}] = \emptyset$, so $\cdot f$ does not preserve meets and has no left adjoint, and the same two subsets give $\bigvee f(\{a_1\} \cup \{a_2\}) = \{b\}$ against $\bigvee f\{a_1\} \cup \bigvee f\{a_2\} = \emptyset$, so $/f^\circ$ does not preserve joins and has no right adjoint. A monic f restores the binary case and no more — τf still reaches only $\text{im } f$, and $\bigvee f\emptyset = B \setminus \text{im } f$ is still not $\emptyset$. The chain extends only when f° is a map as well, and then $/f^\circ = \cdot f^\circ = \cdot f$ and it repeats forever. For an S with neither S nor S° a map there is no triple at all: $\cdot S \dashv / S$ is two links and stops.

The rows that chain forever say nothing by it: $^\circ$ is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and $^\circ \dashv \lrcorner^\circ \dashv \lrcorner^\circ$ has no end, and §1.1's other row of that kind, $\circ \dashv \lrcorner \dashv \lrcorner \circ$, goes the same way.

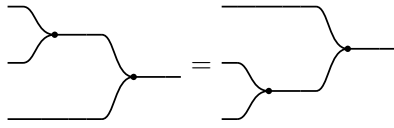
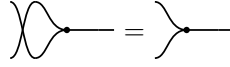
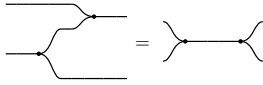
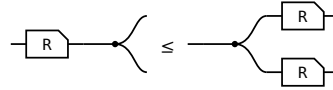
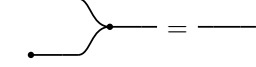
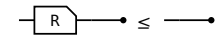
§2. Relations

(2a)

$\mathbf{Rel}$ is a poset-enriched category with $(\otimes, \circ)$ where $\otimes$ is commutative cartesian product, and converse $\circ \dashv \circ$, and

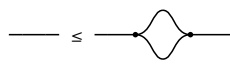
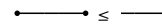
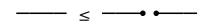
$C: (\triangleright : A \otimes A \rightarrow A, \multimap : I \rightarrow A)$ a commutative monoid with $C^\circ \dashv C$. We write C° as $(\triangleleft : A \rightarrow A \otimes A, \multimap : A \rightarrow I)$

(2b)

 $\triangleright$ associative $\triangleright$ commutative $R \triangleleft \leq \triangleleft (R \otimes R)$  $\multimap$ is its unit $R \multimap \leq \multimap$ Frobenius, one half — the other is its $\circ$

§2.1. $\forall$ object A , $(A, \triangleleft, \multimap) \dashv (A, \triangleright, \multimap)$

(2.1a)

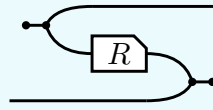
 $\triangleright \triangleleft \leq 1$  $1 \leq \triangleleft \triangleright$  $\multimap \multimap \leq 1$  $1 \leq \multimap \multimap$

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In $\mathbf{Rel}$ it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§3. $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

(3a)

$\circ$ is primitive, part of the data the first section lists. It turns both of R 's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are R° , not its definition:



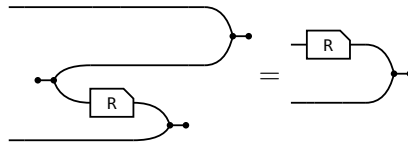
$$R^\circ = (\circ \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \multimap \circ)$$

where $\circ \triangleleft : \mathbb{I} \rightarrow \mathbf{a} \otimes \mathbf{a}$ opens a pair of wires out of nothing and $\triangleright \circ : \mathbf{b} \otimes \mathbf{b} \rightarrow \mathbb{I}$ closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\multimap$ are $\triangleright^\circ$ and $\circ \triangleleft$. The laws of $\circ$ are that it is a contravariant 2-functor $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$:

$$(i) \ 1^\circ = 1 \quad (ii) \ (RS)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:



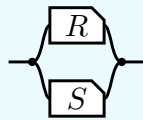
(3.1a)

R° is DEFINED as the bending of $(R \otimes 1) \triangleright \circ$, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the $\circ \triangleleft$ bends the $\mathbf{a}$ strand down and the $\triangleright \circ$ brings it back up, while the $\mathbf{b}$ strand rides through untouched. Nothing else is spent below.

§4. $\cap$ is a commutative idempotent monoid on every hom-set

(4a)

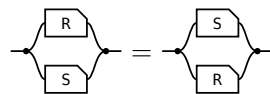
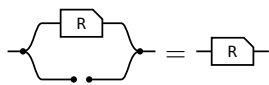
The **meet** of $R, S : a \rightarrow b$, the paper's **convolution**, is $R \cap S := \langle \downarrow (R \otimes S) \rangle$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.



transcribed: a definition has no statement to export

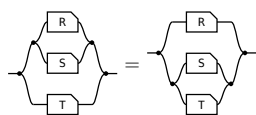
On every hom-set it is associative, commutative and idempotent $\cap$, with unit the maximal arrow $T = \text{---} \bullet \bullet \text{---}$ (the paper's Lemma 4.11).

(4b)

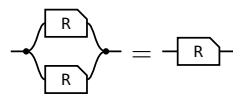


unit: one half of T per end — the merge's unit law absorbs the $\bullet$, the copy's

counit law the $\bullet$



commutative: σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity



associative: coassociativity and associativity; $\otimes$ re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So $\leq$ is the order this monoid induces. $R \cap S \leq R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \cap T$, since $R = R \cap R \leq S \cap T$.

And one law relating $\cap$ to composition, which is **not** an equation:

(4c)

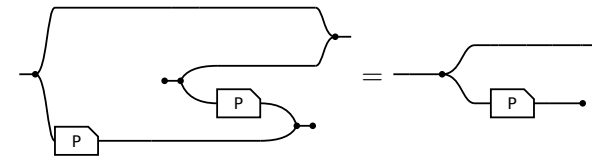
semi-distributivity, and what supplies it	picture
$R \cap (S \cap T) \leq R \cap S \cap T$ — the lax copy law. Equality exactly when R is single valued: the Maps section's $F(R \cap S) = F R \cap F S$.	

§5. Domain and range

(5a)

The **domain** $\text{dom}(R) \triangleq \mathbb{1} \sqcap (R \circ R^\circ)$ and the **range** $\text{ran}(R) \triangleq \text{dom}(R^\circ)$.

(5b)



$\triangleright$ lands the return leg back on the value $\triangleleft$ handed out,
so $R^\circ \triangleright$ cuts to $\circ$ — Frobenius

Running R and throwing the result away leaves only the fact that R could fire, and $\text{ran}(R)$ is the same picture with the box mirrored. In Rel both steps are $\{(a, a) : \exists b. a R b\}$.

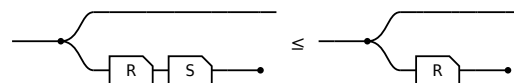
(5c)

1	$\text{dom}(R) \triangleq \mathbb{1} \sqcap (R \circ R^\circ)$
2	$\text{Coreflexive}(X) \implies \text{dom}(R) \sqsubseteq X \iff R \sqsubseteq X \circ R$
3	$\text{dom}(R \circ S) \sqsubseteq \text{dom}(R)$
4	$\text{dom}(R \sqcap S) = \mathbb{1} \sqcap (S \circ R^\circ)$
5	$R \text{ entire} \iff \text{Entire}(R) \triangleq \text{dom}(R) = \mathbb{1}$
6	$R \text{ simple} \iff \text{Simple}(R) \triangleq R^\circ \circ R \sqsubseteq \mathbb{1}$
7	$R \text{ a map} \iff \text{Map}(R) \triangleq \text{Entire}(R) \wedge \text{Simple}(R)$
8	$\text{Entire}(R) \wedge \text{Entire}(S) \implies \text{Entire}(R \circ S)$ $\text{Simple}(f) \wedge \text{Simple}(g) \implies \text{Simple}(f \circ g)$ $\text{Map}(f) \wedge \text{Map}(g) \implies \text{Map}(f \circ g)$
9	$\text{Entire}(R \circ S) \implies \text{Entire}(R)$

§5.1. Sliding the discard

$\text{dom}(RS) \sqsubseteq \text{dom}(R)$, and a single glyph for **dom** would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

(5.1a)



$S \circ \circ \sqsubseteq \circ$, the lax axiom for $\circ$ in the first section
— the discard slides back past S

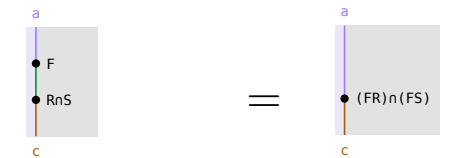
Equality is S entire, which is the same picture read as $\text{dom}(R) = \mathbb{1} \iff R \text{ entire}$.

§6. Maps

(6a)

1	surjective $1 \in R^\circ R$, that is, R° entire.	single valued $R^\circ R \subseteq 1$
2	entire $1 \in RR^\circ$. With single valued, a map.	injective $RR^\circ \subseteq 1$

(6b)

law	picture
$\text{Simple}(F) \implies F(R \cap S) = (F R) \cap (F S)$ F single valued	 one person who admires both a at A, b at B

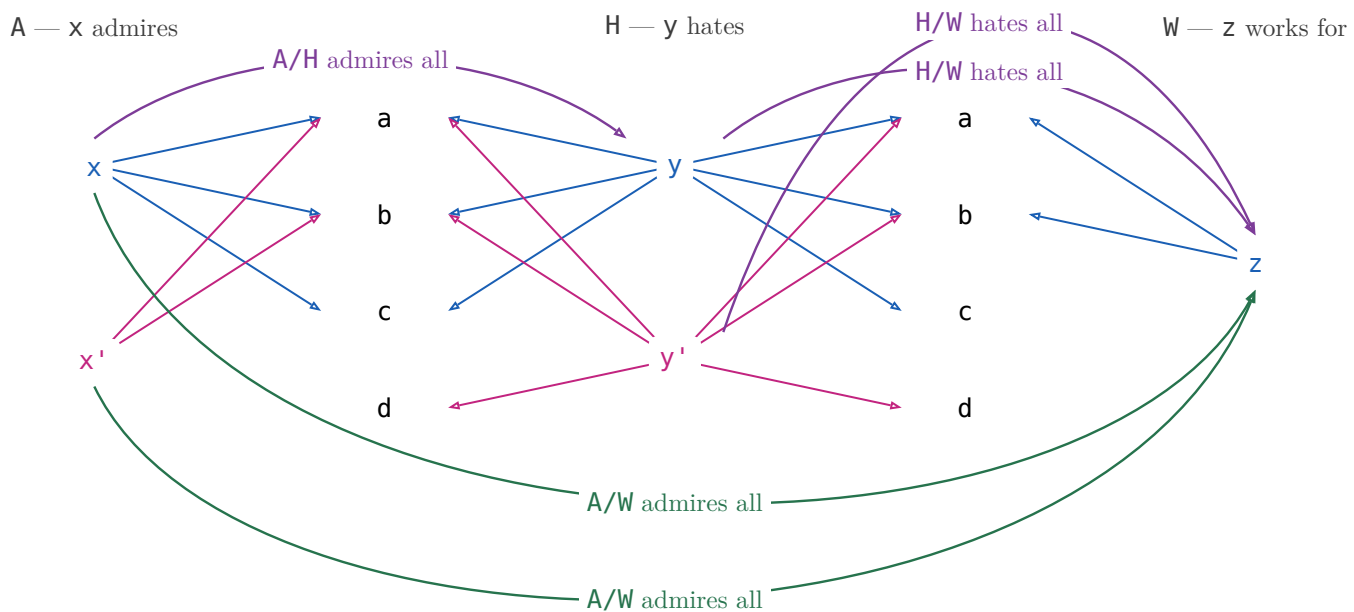
§8. / is all of

(8a)

$x (R/S) y \iff \forall p. y S p \rightarrow x R p$
 $x (S \backslash R) y \iff \forall p. p S x \rightarrow p R y$
 /compares images: $S(y) \subseteq R(x)$. \ compares preimages: $S^\circ(x) \subseteq R^\circ(y)$.
 example: A admires, H hates, W works for
 $x (A/H) y$ — x admires everyone y hates.
 $x (H \backslash A) y$ — everyone who hates x admires y.

§8.1. $(R/S)(S/W) \subseteq R/W$

(8.1a)



(8.1b)

$x (A/H) y$ — x admires everyone y hates
 $y (H/W) z$ — y hates everyone z works for
 $x (A/W) z$ — x admires everyone z works for

$(A/H)(H/W)$ is a path: $x \rightarrow y \rightarrow z$, and that is all of it. A/W also holds of x' , who admires everyone z works for — but x' does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of $(R/S)(S/W) \subseteq R/W$.

1	$X \sqsubseteq R/S \iff XS \sqsubseteq R$ X is any x -to- y pairing; one that only pairs x with a y such that x admires everyone y hates lies inside R/S , and R/S is the largest such.	
2	$X \sqsubseteq S \setminus R \iff SX \sqsubseteq R$ The mirror — divide on the left when x comes first.	
3	$(R/S)S \sqsubseteq R$ There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at $S=\emptyset$: R/S is everyone, $(R/S)S=\emptyset$.	
4	$S (S \setminus R) \sqsubseteq R$ The mirror.	
5	associate: $R/(S_1 S_2) = (R/S_2)/S_1$ A friend's enemy is two hops: divide by the far end first.	
6	$(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$ The mirror.	
7	maps: $f (R/S) = (fR)/S$ Rename x before or after dividing — the licence to write fR/S .	
8	$R/(fS) = (R/S) f^\circ$ Rename y : a map leaves a denominator as f° outside the box.	
9	$(R/S) (S/W) \sqsubseteq R/W$ Someone who admires all of a hate-set that already covers everyone z works for admires those people too.	
10	$1 \sqsubseteq R/R$ R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
11	$(R/R) (R/R) = R/R$ R/R is the preorder admires at least as much as , and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $1 \sqsubseteq R/R$ above it is an equality.	
12	$(R/R)R = R$ Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.	
13	$R/1 = R$ Dividing by 1 : p 's set is just $\{p\}$, so admiring all of it is admiring p .	
14	$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ Admiring a combined hate-set is admiring each set in full.	
15	$S \setminus (R/W) = (S \setminus R)/W$ Which is why $S \setminus R/W$ needs no bracket.	

Fifteen laws, fifteen pictures, and not one shows a generator: η , υ , $\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.

§9. $\frac{R}{S}$

(9a)

$R /_S S \triangleq (R / S) \cap (S / R)^\circ$ what R sends x to and nothing else is what S sends y to. In Rel x and y has the same image: $\forall p. (x R p \iff y S p)$

(9b)

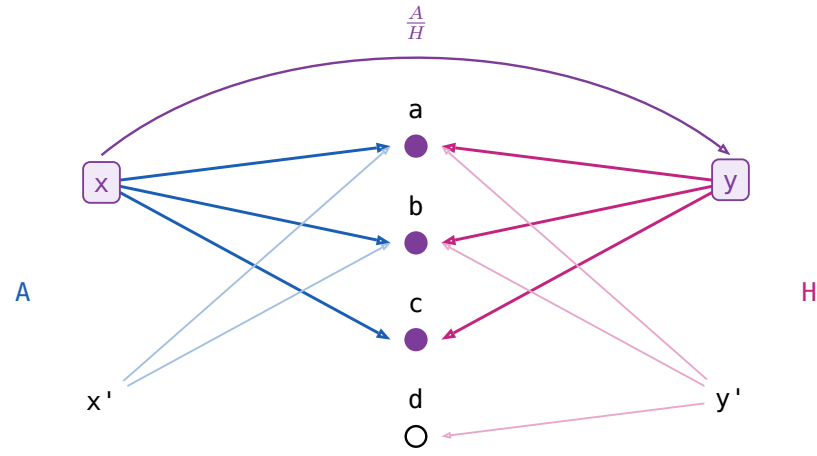
$x (A/H) y$ — x admires everyone y hates

$x ((H/A)^\circ) y$ — x admires only people y hates

$x ((A/H) \cap (H/A)^\circ) y$ — x admires only and all whom y hates

x admires exactly whom y hates: $/$ is **all of**, $\frac{R}{S}$ is **only all of**.

(9c)



	law	the reading
1	$X \sqsubseteq R /_s S \iff X S \sqsubseteq R \wedge X^\circ R \sqsubseteq S$ X sits inside the matching exactly when it carries S back inside R and, reversed, carries R back inside S.	X may pair x with y only when x admires exactly whom y hates. Both halves must typecheck, so the operation is partial .
2	$(R /_s S)^\circ = S /_s R$ reversing the matching swaps the two relations.	Matching is symmetric.
3	$(R /_s S) (S /_s W) \sqsubseteq R /_s W$ a match from R to S followed by one from S to W is a match from R to W.	And transitive.
4	$(R /_s S) S \sqsubseteq R$ following the matching by S lands inside R.	$(\exists y. x(\frac{R}{S})y \wedge ySp) \rightarrow xRp$ x only admires whom y hates $\frac{R}{S}S = \text{Dom}(\frac{R}{S})R$
5	$(R /_s R) R = R$ matching R against itself and then following R gives R back.	$(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$ x and y admire the same people y=x always qualifies ($1 \sqsubseteq R\%R$ below)
6	$1 \sqsubseteq R /_s R$ every x matches itself.	$x(\frac{R}{R})y$ if x and y admires the same peoples.
7	$(R /_s R) (R /_s R) = R /_s R$ matching twice matches no more pairs than matching once.	So the relation admires the same people is an equivalence relation.
8	$\text{Symmetric}(X) \implies X \sqsubseteq R /_s R \iff X R \sqsubseteq R$ a symmetric X sits inside the matching exactly when following it by R adds nothing to R.	The largest symmetric arrow that leaves R alone.
9	$\text{simplePart}(R) \triangleq R /_s 1$ matching R against the identity keeps the X whose image is a single point.	The people who admire exactly one person and nobody else. It equals R only when R is simple, unlike $R/1=R$.
10	$\text{dom}(R /_s S) = 1 \cap ((R / S) (S / R))$ the X that match something are those the two one-sided divisions take round the loop back to X.	Its domain is the domain of simplicity of R.

§10. Freyd's Power allegories

(10a)

A **power allegory** is a division allegory with one unary operation on arrows, $\exists$ **epsilon**, subject to

- 1 $\exists_R \square = R\square, \quad \exists_R = \exists_R \square$
- 2 $\mathbb{1} \sqsubseteq (R/\exists_R)(\exists_R/R) \quad \exists_R \text{ is **thick**}$
- 3 $\frac{\exists_R}{\exists_R} = \mathbb{1} \quad \exists_R \text{ is **straight**}$

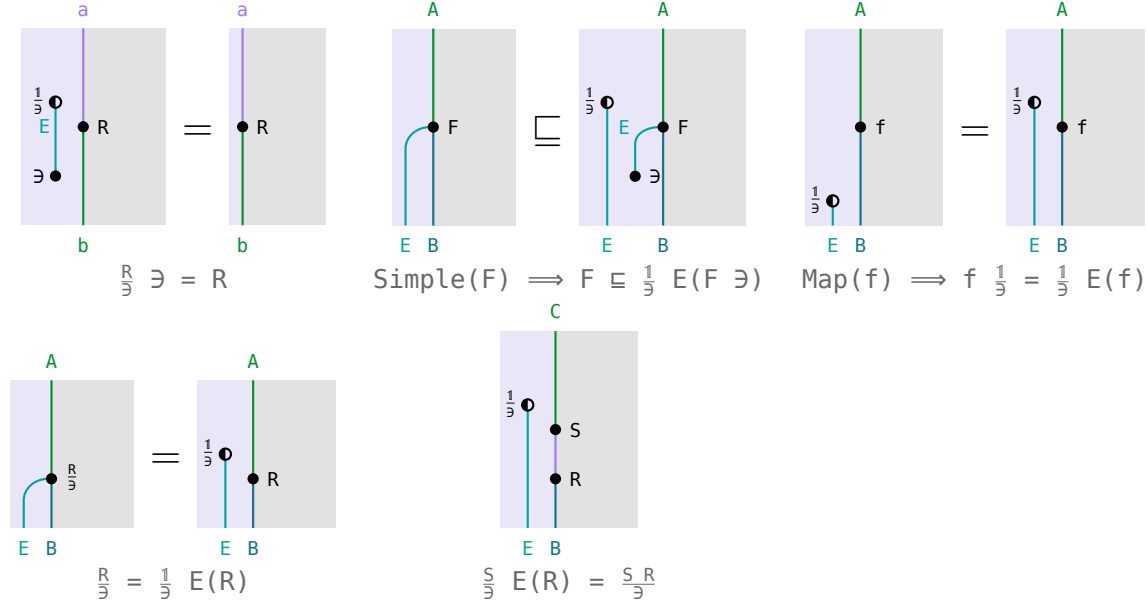
$R\square$ is R 's target, an identity arrow. For $R : A \rightarrow B$ write $\exists : PB \rightarrow B$, dropping the subscript.

$\epsilon \triangleq \exists^\circ : A \rightarrow PA$

(10b)

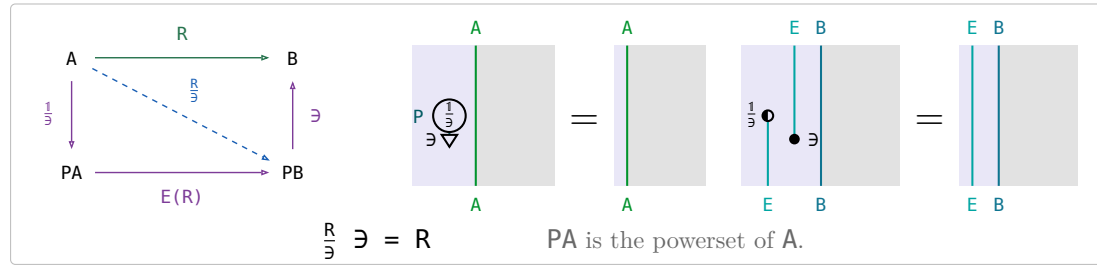
	law	the reading
1	$\exists_R \square = R\square, \quad \exists_R = \exists_R \square$	One $\exists$ per object, not per arrow.
2	$\exists$ is thick	Comprehension: every x has a set of exactly the people x admires.
3	$\frac{R}{\exists} : A \rightarrow PB$, for $R : A \rightarrow B$	convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$
4	$\text{Map}(\frac{R}{\exists})$	
5	$\frac{R}{\exists} \exists = R$	reading the chosen set back through $\exists$ returns the relation.
6	$\text{Simple}(F) \implies F \sqsubseteq \frac{F \exists}{\exists}$	A partial choice of sets is inside the total one.
7	the singleton map is monic: $\frac{1}{\exists} (\frac{1}{\exists})^\circ \sqsubseteq \mathbb{1}$ two points with the same one-person set are the same point.	The one-person set.
8	$\frac{\exists}{\exists} = \mathbb{1}$	Make the set of a set, then read it back one level down.
9	fusion: $\text{Map}(f) \implies \frac{f R}{\exists} = f \frac{R}{\exists}$	Naturality of the unit, $\text{Map}(f) \implies f \frac{1}{\exists} = \frac{1}{\exists} E(f)$. renaming a point and then taking its one-person set is taking the set and renaming inside it.
10	$\text{Map}(f) \wedge \mathbb{1}\square = \exists\square \implies \frac{f}{\exists} = f \frac{1}{\exists}$	Rename first or take singletons first — the fusion row above at $R=1$.
11	$R\square = \exists\square \implies R /_s S = \frac{R}{\exists} (\frac{S}{\exists})^\circ$	x and y match when R sends x and S sends y to the same set.
12	$E(R) \triangleq \frac{\exists R}{\exists}$	$E(R) : PA \rightarrow PB$, image of a set of A
13	$\frac{R}{\exists} = \frac{1}{\exists} E(R)$	$\frac{1}{\exists} : x \mapsto \{x\}$
14	$\frac{S}{\exists} E(R) = \frac{S R}{\exists}$	absorption — the monad's composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{S R}{\exists}$, §10.3
15	$E(R) = E(\frac{R}{\exists})$ union	the image of a set under R is the union of the R -sets of its points.
16	$\supseteq \triangleq \exists / \exists$	$xs \supseteq ys \iff \forall a. ys \exists a \rightarrow xs \exists a$
17	$\subseteq = \supseteq^\circ$	$xs \subseteq ys \iff ys \supseteq xs$
18	$T^\circ \frac{T}{\exists} \sqsubseteq \epsilon$	every T -value of a point is a member of its T -set: $\wedge(T)$ is a map and $\wedge(T)\exists = T$.

(10c)



§10.1. $\mathbf{i}\text{-E}$ Power Allegory defined as adjunction

(10.1a)



$\mathbf{i}$ is the inclusion $\text{Map}(\mathcal{A}) \rightarrow \mathcal{A}$, doing nothing, so E is both the functor and the monad $\mathbf{i}E$.

§10.2. $E \setminus$ as a composite

$$E \setminus Z = \subseteq \left(\frac{Z^\circ}{\exists} \right)^\circ \quad (10.2a)$$

xs is related to c by $E \setminus Z$ exactly when $xs \subseteq \Lambda(Z^\circ)(c)$, the set of every a with $a Z c$

(a)

(b)

(c)

(a)	$E \setminus Z$	$Z = E \Lambda(Z^\circ)^\circ \text{ — } \frac{R}{\exists} \exists = R$
(b)	$= E \setminus (E (\frac{Z^\circ}{\exists})^\circ)$	$X \setminus (Y f^\circ) = (X \setminus Y) f^\circ, f \text{ a map}$
(c)	$= \subseteq (\frac{Z^\circ}{\exists})^\circ$	

$$R \setminus S = \frac{R^\circ}{\exists} \subseteq \left(\frac{S^\circ}{\exists} \right)^\circ \quad (10.2b)$$

b is related to c by $R \setminus S$ exactly when $\Lambda(R^\circ)(b) \subseteq \Lambda(S^\circ)(c)$, the R -preimage of b inside the S -preimage of c

(a)

(b)

(c)

(d)

(a)	$R \setminus S$	$R = E \Lambda(R^\circ)^\circ \text{ — } \frac{R}{\exists} \exists = R$
(b)	$= (E (\frac{R^\circ}{\exists})^\circ) \setminus S$	$(XY) \setminus S = Y \setminus (X \setminus S)$
(c)	$= (\frac{R^\circ}{\exists})^\circ \setminus (E \setminus S)$	$f^\circ \setminus X = fX; E \setminus S = \subseteq \Lambda(S^\circ)^\circ \text{ — } E \setminus Z = \subseteq (\frac{Z^\circ}{\exists})^\circ$
(d)	$= \frac{R^\circ}{\exists} \subseteq (\frac{S^\circ}{\exists})^\circ$	

§10.3. $\mathcal{A} \cong \mathbf{Kleisli}(E)$

(10.3a)

$$E(R) \triangleq \frac{\exists}{\exists} R, \quad \frac{1}{\exists} : A \rightarrow PA, \quad \text{union} = E(\exists) : PPA \rightarrow PA$$

$$f \diamond g \triangleq f \circ E(g) \circ \text{union}, \quad \text{for } f : A \rightarrow PB \text{ and } g : B \rightarrow PC$$

the monad is on $\mathbf{Map}(\mathcal{A})$, not on the allegory: E is a relator on all relations, but $\frac{1}{\exists}$, union and $f \circ E(g) \circ \text{union}$ are maps, and the Kleisli construction happens where they live.

(10.3b)

$$\begin{aligned} \boxed{\frac{S}{\exists} \diamond \frac{R}{\exists}} & \xrightarrow{\text{definition of } \diamond, \text{ union}=E(\exists)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\right) E(\exists)} \xrightarrow{E \text{ a functor, } E(X)E(Y)=E(XY)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\exists\right)} \\ & \xrightarrow{\frac{R}{\exists} \exists = R, \frac{R}{\exists} \exists = R} \boxed{\frac{S}{\exists} E(R)} \xrightarrow{\frac{S}{\exists} E(R) = \frac{S}{\exists} R, \text{ absorption}} \boxed{\frac{SR}{\exists}} \end{aligned}$$

the first three steps only unfold $\diamond$ — E is a functor, so $E\left(\frac{R}{\exists}\right)\text{union}=E\left(\frac{R}{\exists}\exists\right)=E(R)$ and the union is gone.

Absorption, $\frac{S}{\exists} E(R) = \frac{S}{\exists} R$, is the whole law, and it is the functoriality of $\frac{\cdot}{\exists}$.

1 goes to $\frac{1}{\exists}$, the Kleisli identity, by definition — so $\frac{\cdot}{\exists}$ is an isomorphism of categories $\mathcal{A} \cong \mathbf{Kleisli}(E)$, and §10.1's $i \dashv E$ is the Kleisli adjunction. i is the identity on objects, so every object of the allegory is free.